

2f. Detailed Content of A Level Mathematics A (H240)

1 – Pure Mathematics

When this course is being co-taught with AS Level Mathematics A (H230) the ‘Stage 1’ column indicates the common content between the two specifications and the ‘Stage 2’ column indicates content which is particular to this specification.

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.01 Proof				
1.01a 1.01d	Proof	a) Understand and be able to use the structure of mathematical proof, proceeding from given assumptions through a series of logical steps to a conclusion. <i>In particular, learners should use methods of proof including proof by deduction and proof by exhaustion.</i>	d) Understand and be able to use proof by contradiction. <i>In particular, learners should understand a proof of the irrationality of $\sqrt{2}$ and the infinity of primes.</i> <i>Questions requiring proof by contradiction will be set on content with which the learner is expected to be familiar e.g. through study of GCSE (9–1), AS or A Level Mathematics.</i>	MA1
1.01b		b) Understand and be able to use the logical connectives $\equiv$, $\Rightarrow$, $\Leftrightarrow$. <i>Learners should be familiar with the language associated with the logical connectives: “congruence”, “if..... then” and “if and only if” (or “iff”).</i>		
1.01c		c) Be able to show disproof by counter example. <i>Learners should understand that this means that, given a statement of the form “if $P(x)$ is true then $Q(x)$ is true”, finding a single x for which $P(x)$ is true but $Q(x)$ is false is to offer a disproof by counter example.</i> <i>Questions requiring proof will be set on content with which the learner is expected to be familiar e.g. through study of GCSE (9–1) or AS Level Mathematics.</i> <i>Learners are expected to understand and be able to use terms such as “integer”, “real”, “rational” and “irrational”.</i>		